



DOBOT

User Guide

Force Sensor

User Guide

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Shenzhen Yuejiang Technology Co., Ltd.

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The user has the responsibility to make sure of following the relevant practical laws and regulations of the country, in order that there is no significant danger in the use of the robot.

Shenzhen Yuejiang Technology Co., Ltd.

Address: Address: Floor 9-10, Building 2, Chongwen Garden, Nanshan iPark, Liuxian Blvd,
Nanshan District, Shenzhen, Guangdong Province, China

Website: www.dobot.cc

Preface

Purpose

This manual introduces the basic performance, installation and operation of force sensor, which is convenient for users to understand and use the force sensors.

Intended Audience

This document is intended for:

- Customer
- Sales Engineer
- Installation and Commissioning Engineer
- Technical Support Engineer

Change History

Date	Change Description
2022/7/29	The first release

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
 DANGER	Indicates a hazard with a high level of risk which, if not avoided, could result in death or serious injury
 WARNING	Indicates a hazard with a medium level or low level of risk which, if not avoided, could result in minor or moderate injury, robot damage
 NOTICE	Indicates a potentially hazardous situation which, if not avoided, can result in equipment damage, data loss, or unanticipated result
 NOTE	Provides additional information to emphasize or supplement important points in the main text

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1. Description on Sensor

1.1 KWR75 series

KWR75 series sensor is a strain type six-axis force/torque sensor. It can simultaneously detect three forces and three torques. The sensor detects the relative deformations between the tool-end flange and the main body caused by applied forces, and uses a resistance strain gauge to measure changes in the elastic element of the sensor. The sensor contains an embedded system, which can collect and process the signal changes of the resistance strain gauge in real time, and output the magnitude and direction of the applied force in real time, with high precision and response ability.



Figure 1.1 KWR75 sensor

1.2 ATI series

ATI series sensor is a strain type six-axis force/torque sensor. It can simultaneously detect three forces and three torques. The sensor has three indicator lights: L/A, DIAG and STATUS, which can directly acquire the current status of the sensor.



Figure 1.2 ATI sensor

1.3 XJC-3F series

XJC-3F series sensor is a strain type three-dimensional force sensor. It can simultaneously detect three forces. With mechanical anti-overload design, the sensor has high anti-overload ability. The integrated amplifier can be built in the sensor, and the output signal has a variety of forms. Through a special decoupling method, the crosstalk error of the sensor is very small, especially the long-term crosstalk error.



Figure 1.3 XJC-3F sensor

2. Installation Instruction

2.1 Mechanical installation

2.1.1 KWR75 series

Step 1 Check whether the surface of the end of robot arm, connecting flange and sensor are damaged or covered.

Step 2 Separate the connecting flange from the sensor, and install the connecting flange at the end of the robot arm.

Unscrew the six embedded hex socket screws using a 3mm inner-hexagon wrench to separate the connecting flange from the sensor.

Position the connecting flange and the end of robot arm through a $\phi 6$ locating pin. Fix the connecting flange and the end of robot arm using four M6 inner hexagon screws.

Step 3 Fix the sensor to the connecting flange.

Position the sensor with the connecting flange using the locating pin.

Tighten the sensor using six embedded screws. Insert an inner hexagon wrench (width: 3mm) into the mounting hole of the sensor tool end flange and turn it in the direction of the right-handed screw to secure. Please follow the sequence in Figure 2.2 to tighten the screws.

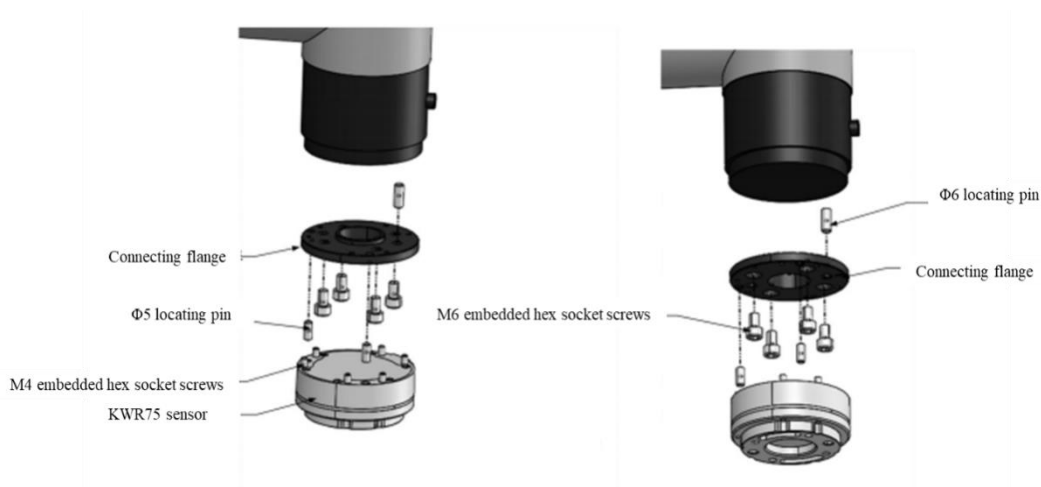


Figure 2.1 KWR75 sensor installation diagram

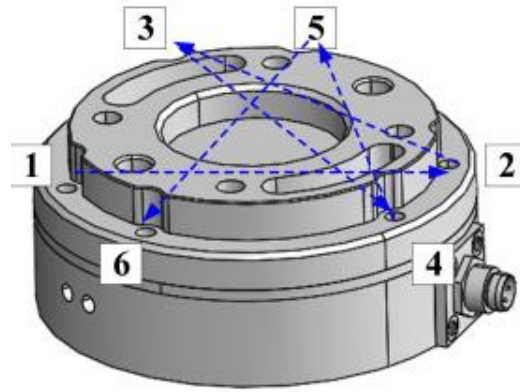


Figure 2.2 Sequence of screwing

Step 4 Connect the tool with the tool flange of the sensor.

The tool flange of sensor provides four M6 screw holes and $\phi 6$ pin holes for connecting tools.

2.1.2 ATI series

The mechanical installation procedure is the same as KWR75 series, as shown in Figure 2.3.

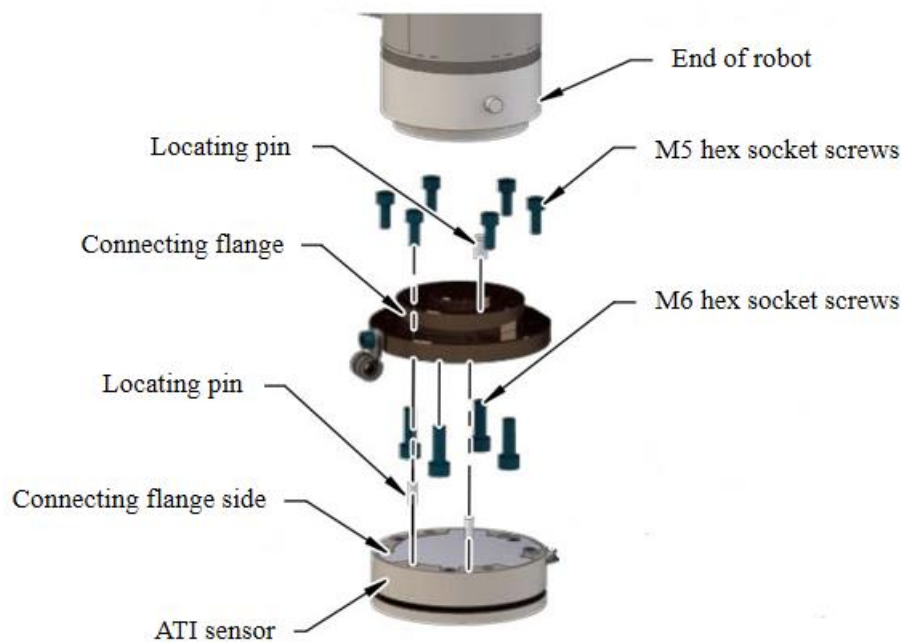


Figure 2.3 ATI sensor installation diagram

2.1.3 XJC-3F series

The mechanical installation procedure is the same as KWR75 series.



The flange locating pin on the tool end of the sensor is used for the consistency of the tool installation. If the locating pin is not used, the sensor performance will not be affected.

2.2 Electrical connection

2.2.1 KWR75 series

The figure below shows the connection between CR robot and KWR75 sensor. You should plug the KWR75 cable to the CR I/O interface following the right cable sequence.

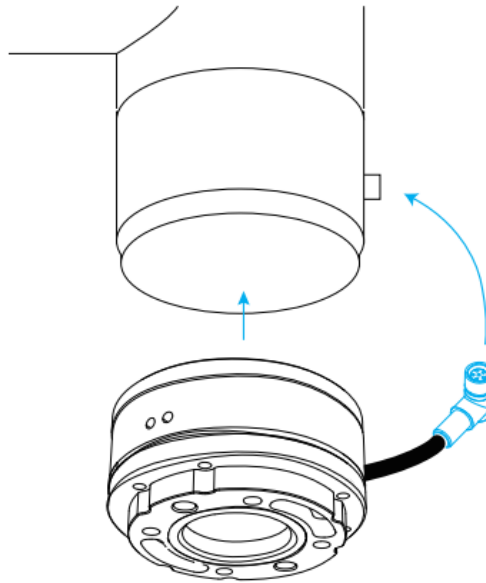


Figure 2.4 Electrical connection

2.2.2 ATI series

The electrical connection procedure is the same as KWR75 series.

2.2.3 XJC-3F series

Electrical interface description:

The color of XJC-3F sensor cable is shown in Figure 2.5, and described in Table 2.1.

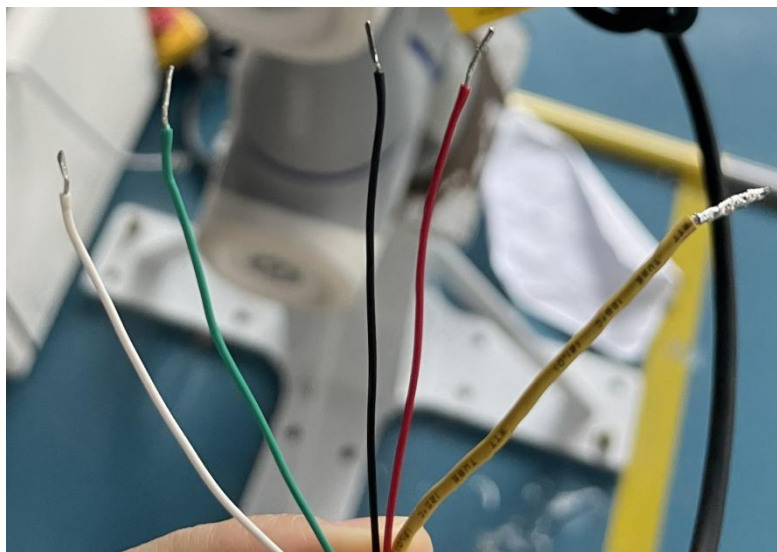


Figure 2.5 XJC-3F sensor cable

Table 2.1 XJC-3F sensor cable definition

Color	Definition
White	RS485 B
Green	RS485 A
Red	24V +
Black	24V -
Yellow	GND

The I/O extension cable of the robot end and the definition of end I/O is shown in Figure 2.6.

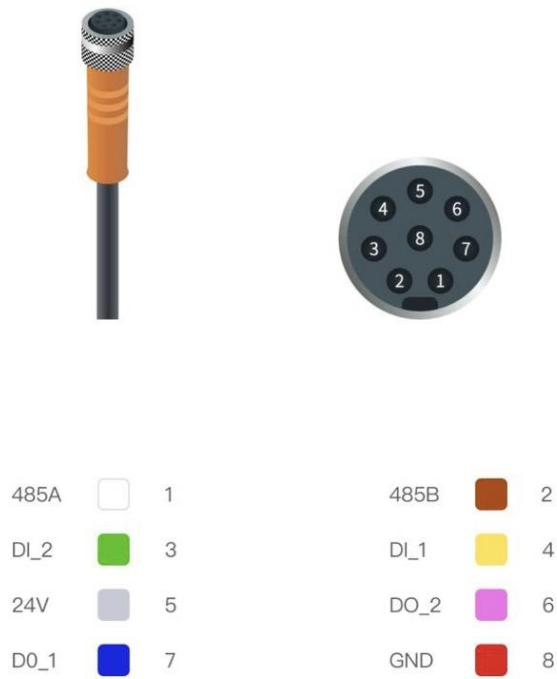


Figure 2.6 I/O definition of the robot end

Electrical connection

Connect the sensor cable to the corresponding I/O extension cable according to the cable definition. Then plug the I/O cable to the I/O interface of the robot end following the right cable sequence, as shown in Figure 2.7.

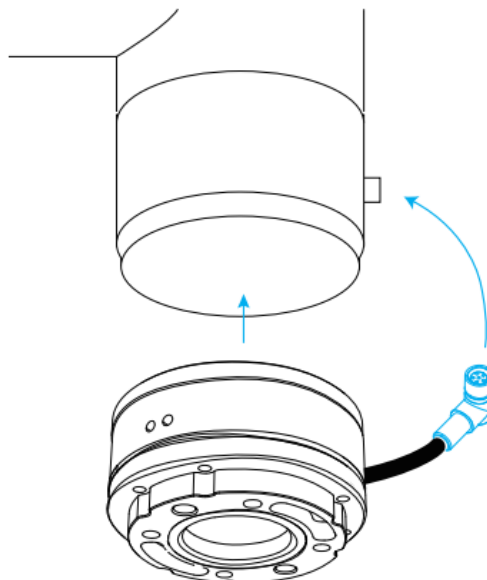


Figure 2.7 Electrical connection

3. Installation of Plug-in

The sensor plug-in needs to be installed through CR software. For Windows, please use DobotSCStudio V2.2.1 or later versions. For Android, please use Android CRStudioV4.0.0.6 or later versions. For iOS, please use iOS CRStudio V2.1.0 or later versions.

3.1 Installation in Windows

This section mainly introduces how to install the sensor plug-in DobotSCStudio software in Windows.

3.1.1 Modify ID configuration

Before connecting the robot, you need to modify the ID configuration of ATI series sensors and then operate in the upper computer. For KWR75 series and XJC-3F series, you do not need to modify the ID configurations (“sensorType” is 10 by default). See *3.1.2 Operations* directly for details.

Step 1 Update the robot controller to ATI test version, as shown in Figure 3.1.

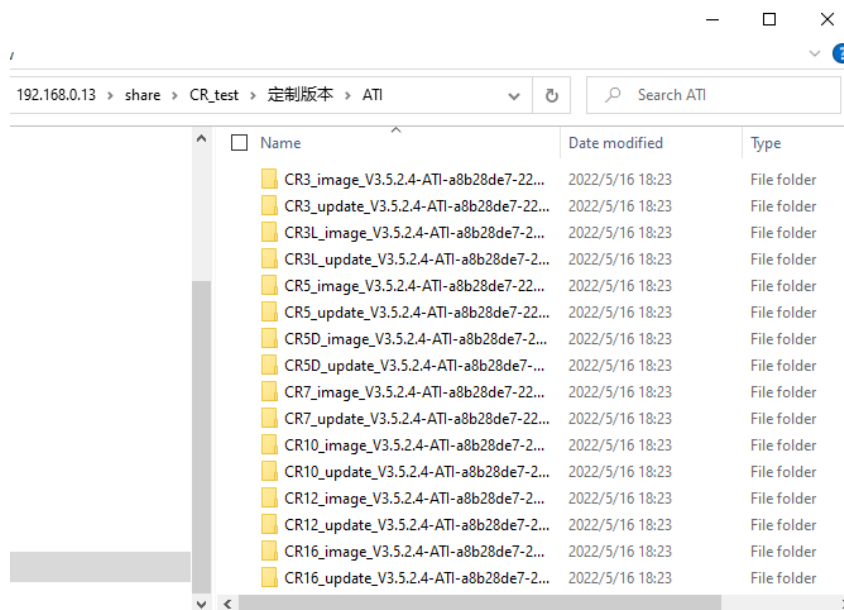


Figure 3.1 Update controller

Step 2 Enter the controller folder, as shown in Figure 3.2, and open the “config” file.

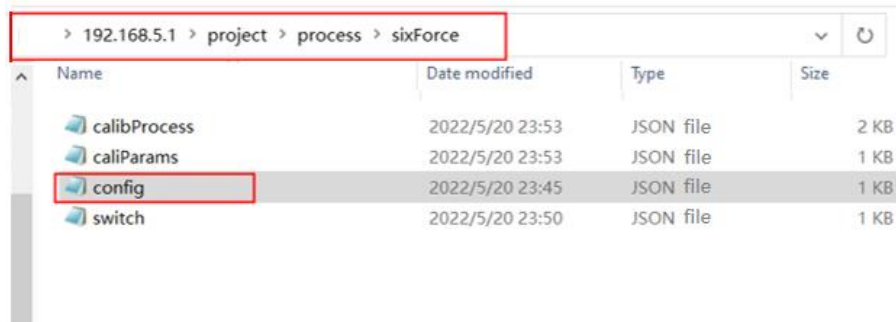


Figure 3.2 Open the force sensor file

Step 3 Modify the **sensorType** of the ATI sensor to **14**, as shown in Figure 3.3.

```
config - 记事本
文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)
{
  "baud": 115200,
  "sensorType": 14
}
```

Figure 3.3 Modify ID of ATI sensor

Step 4 After the modification, power off the robot arm and restart it.

3.1.2 Operations

Step 1 After opening DobotSCStudio and connecting the robot, click **System > Dobot+**, as shown in Figure 3.4.

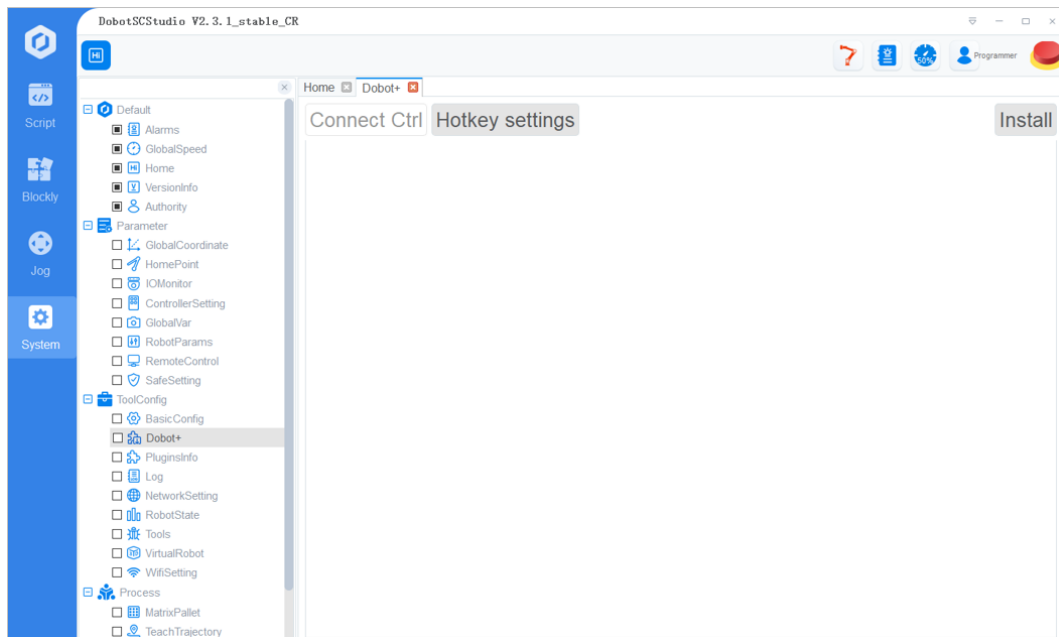


Figure 3.4 Dobot+ page

Step 2 Click **Install** in the “Dobot+” page.

Step 3 Double-click **ForceTorqueSensor.zip**. to install it, as shown in Figure 3.5.

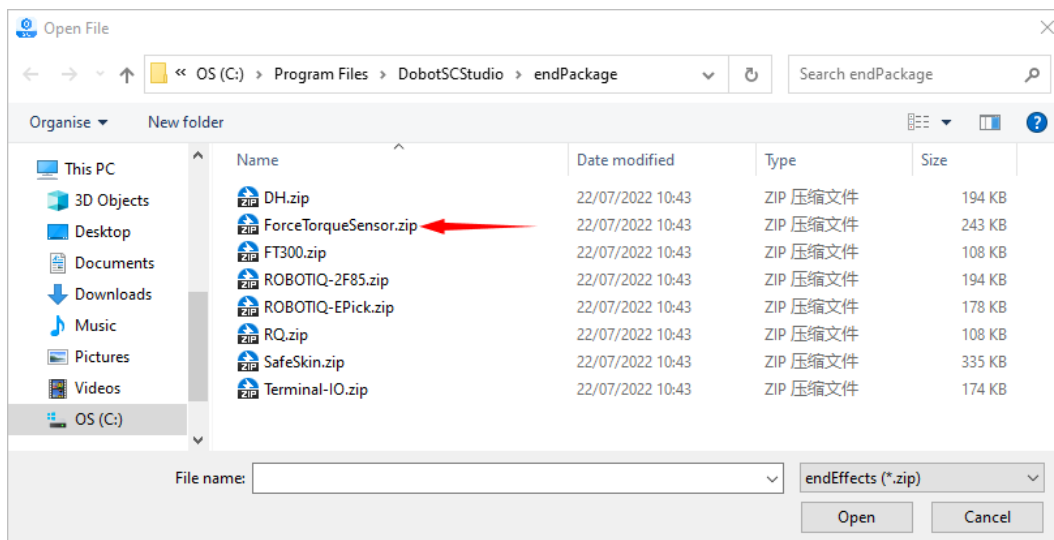


Figure 3.5 Install ForceTorqueSensor.zip

Step 4 After installation, the Panel tab is displayed on the screen, as shown in Figure 3.6. The parameters in Panel tab are described in Table 3.1.

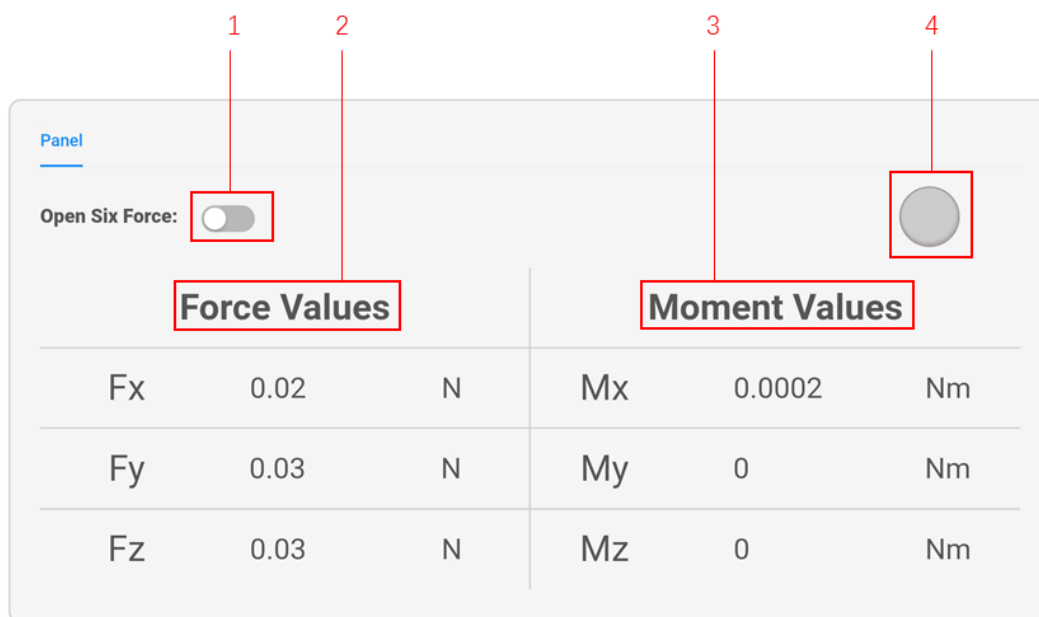


Figure 3.6 Panel tab

Table 3.1 Panel parameters

No.	Description
1	Open/close six force
2	Force value
3	Moment value

4	Status of sensor When the sensor is turned on and in online status, the icon will turn green
---	---

Step 5 Enable **Open Six Force**, as shown in Figure 3.7. The Teach and Setting tabs are displayed on the page. You can reset and calibrate the sensor in Setting tab, and adjust the dragging mode and speed in Teach tab. For detailed operations, see *4Sensor Calibration and Functions*.

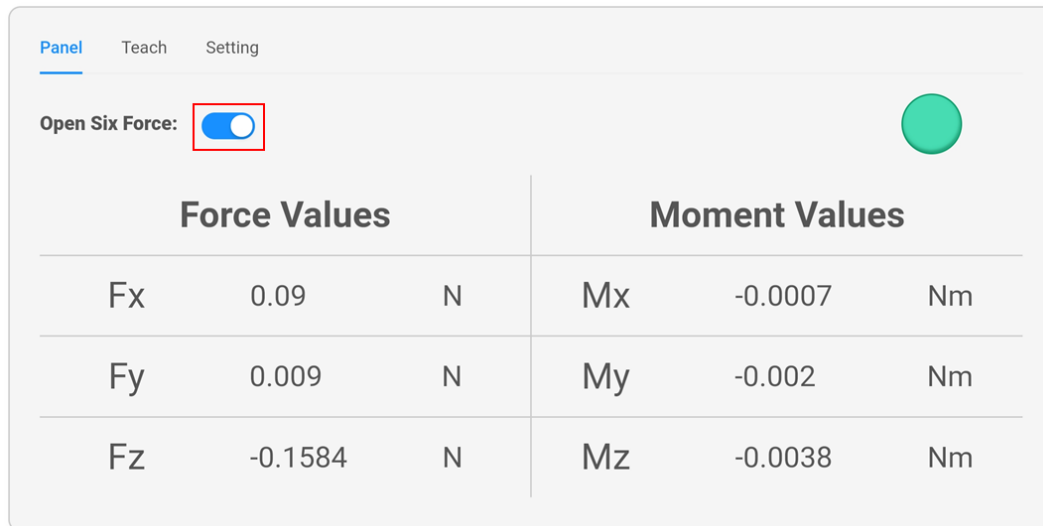


Figure 3.7 Open six force

3.2 Installation in Android & iOS

Prerequisites

Select the right WiFi of the robot and connect it. The default WiFi name of CR robots is Dobot_WIFI_xxx and the default password is 1234567890. After connecting WiFi, if the “This WLAN network has no Internet access. Connect anyway?” window pops up, click **Connect**.

Applicable products

KWR75 series and XJC-3F series sensors

Procedure

Step 1 Click **Monitor** or  to enter the monitor page. Select **Dobot+** in the left-side menu, as shown in Figure 3.8.

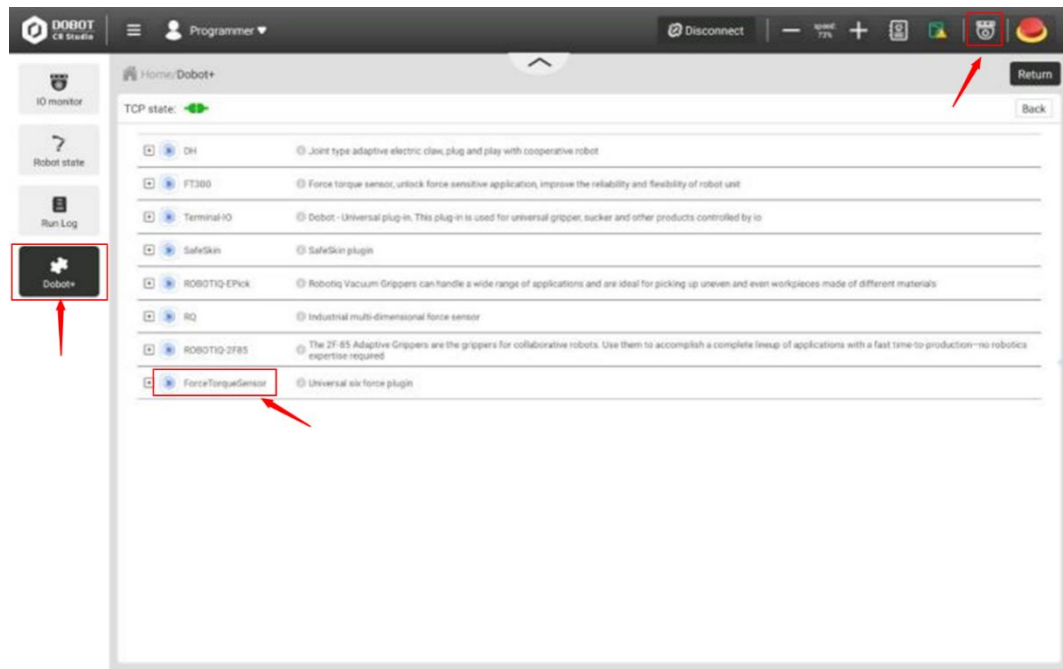


Figure 3.8 Install ForceTorqueSensor plug-in

Step 2 Double-click the **ForceTorqueSensor** plug-in to install it. Then the Panel tab will display on the screen, as shown in Figure 3.9. The parameters in Panel tab are described in Table 3.2.

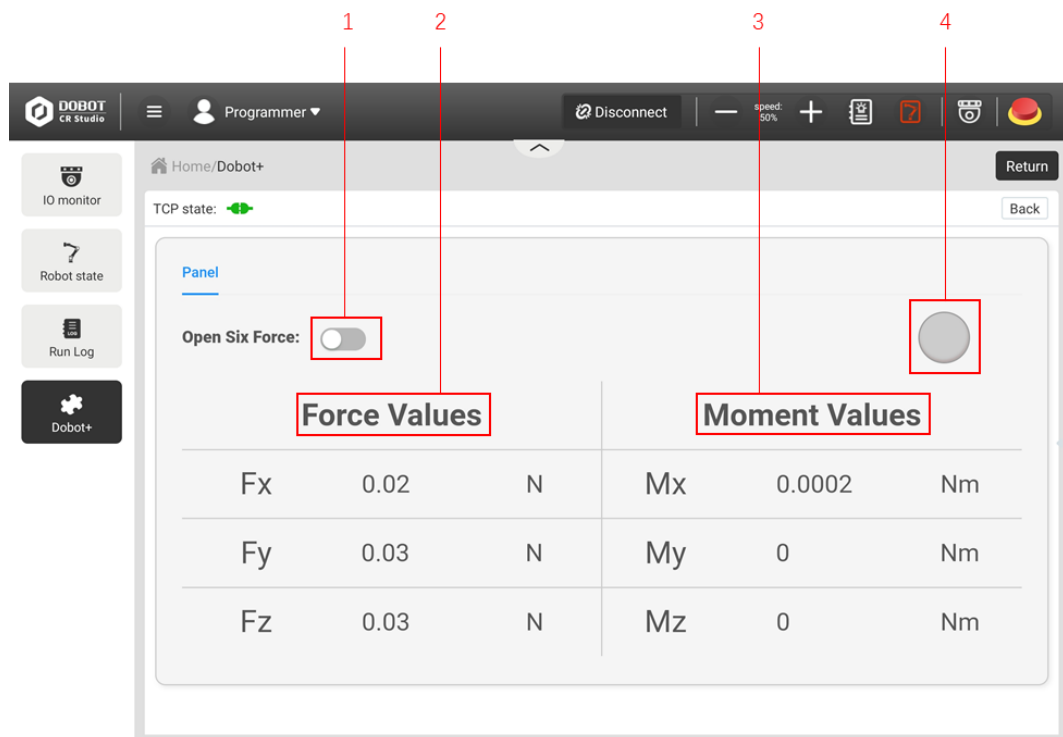


Figure 3.9 Panel page

Table 3.2 Description on Panel parameters

No.	Description
-----	-------------

1	Open/close six force
2	Force value
3	Moment value
4	Status of sensor When the sensor is turned on and in online status, the icon will turn green

Step 3 Enable **Open Six Force**, as shown in Figure 3.10. The Teach and Setting tabs are displayed on the page. You can reset and calibrate the sensor in Setting tab. You can adjust the dragging mode and speed in Teach tab. For detailed operations, see *4Sensor Calibration and Functions*.

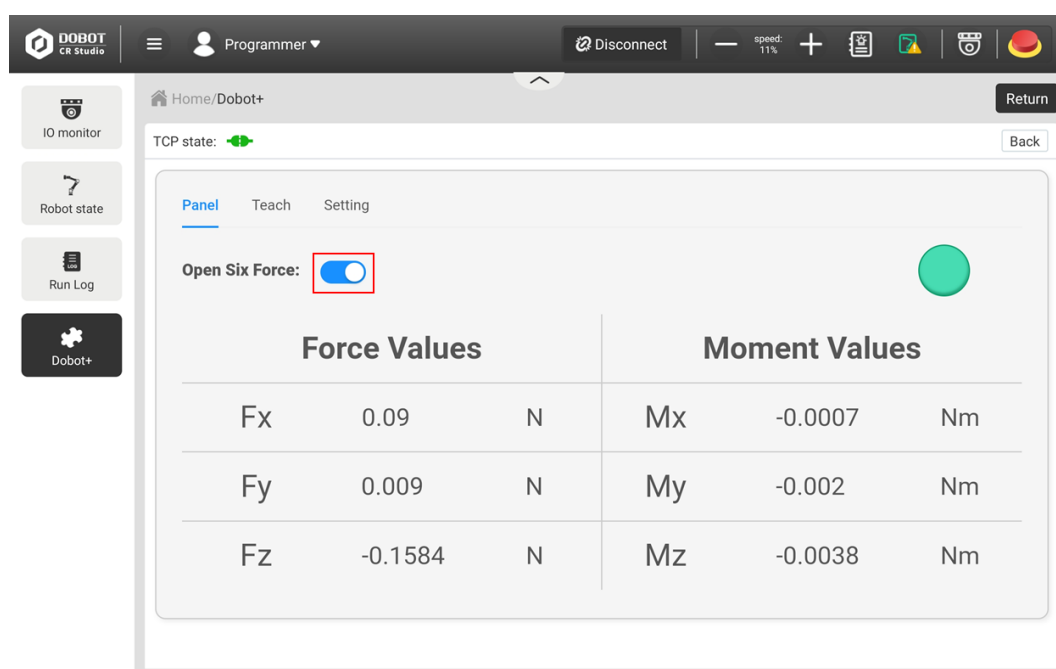


Figure 3.10 Open six force

4. Sensor Calibration and Functions

4.1 Sensor calibration

- Step 1** Jog or drag the robot to an appropriate position (such as $J1=0^\circ$, $J2=0^\circ$, $J3=90^\circ$, $J4=90^\circ$, $J5=90^\circ$), and position the X+ at the bottom of the sensor.

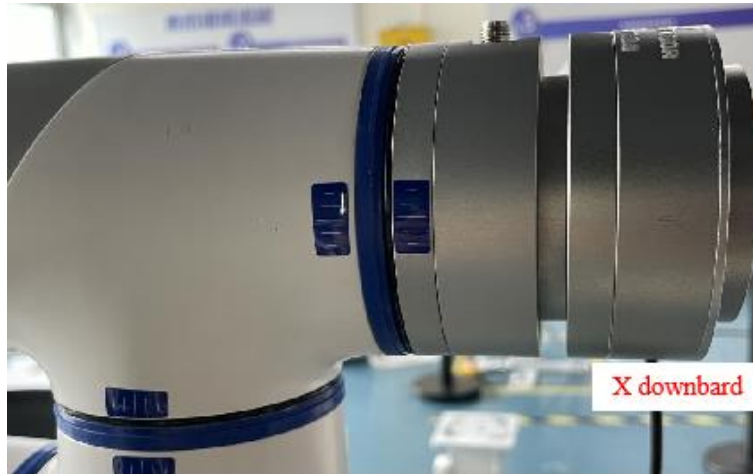


Figure 4.1 Position of X+

- Step 2** Enter the Setting page. Click **Continue** to calibrate the X-axis direction.

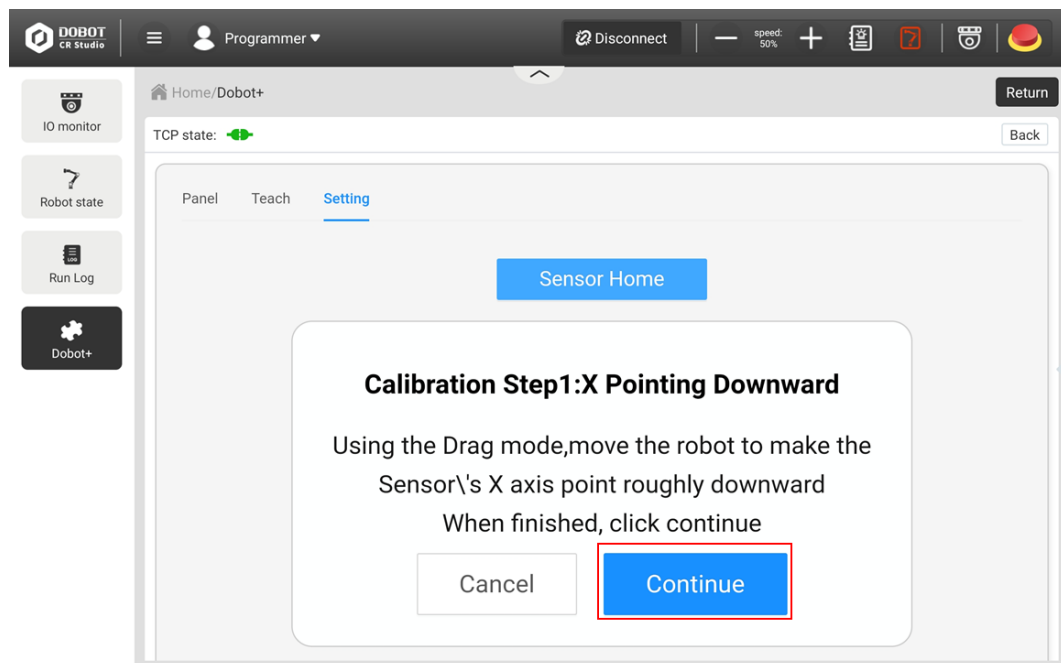


Figure 4.2 Calibrate X-axis direction

- Step 3** Jog or drag the robot to make the Y+ at the bottom of the sensor, as shown in Figure 4.3.

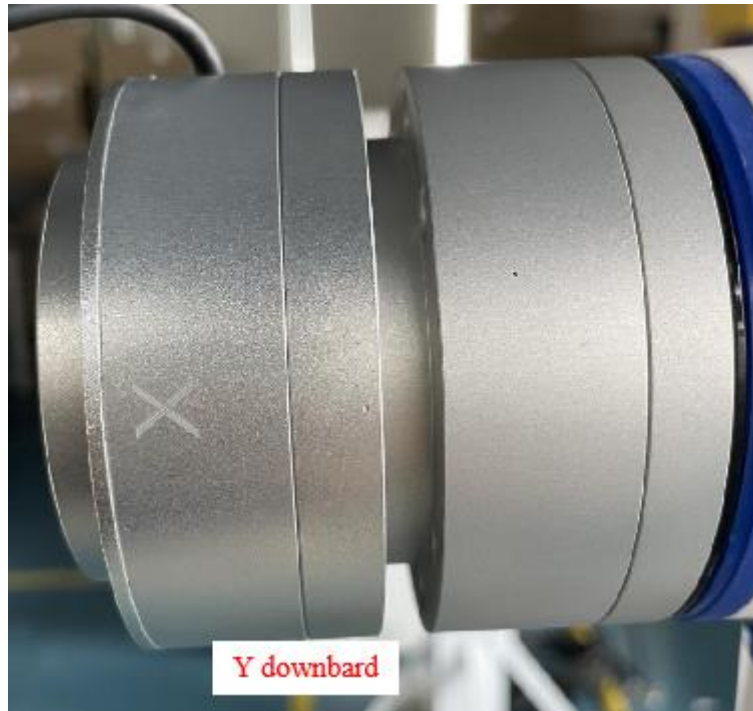


Figure 4.3 Y+ at the bottom

Step 4 Click **Continue** in the Setting page to calibrate the Y-axis direction, as shown in Figure 4.4.

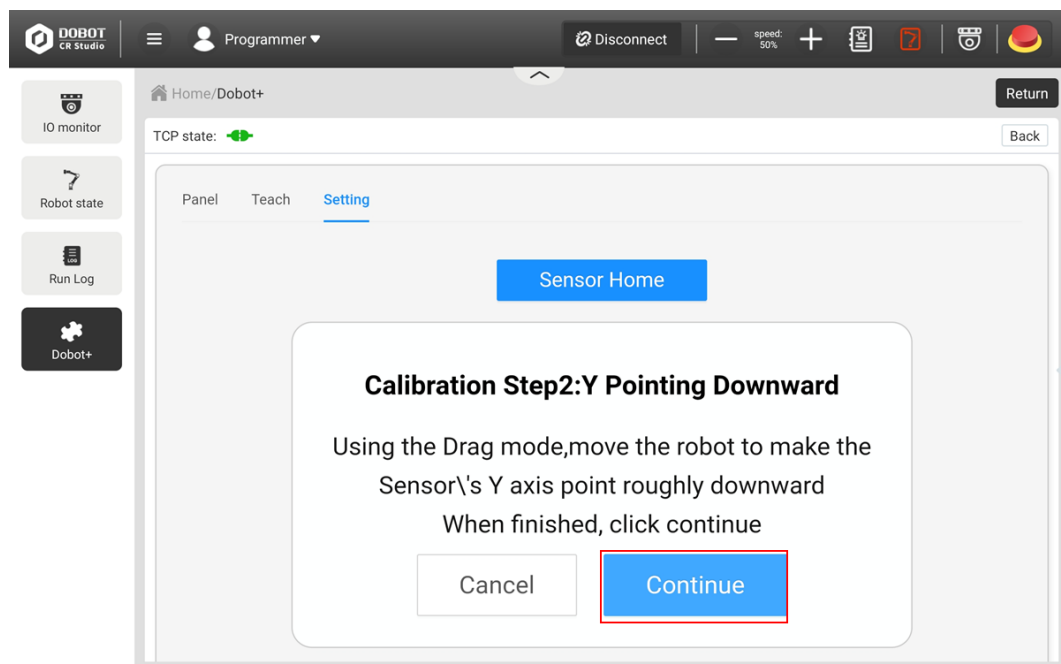


Figure 4.4 Calibrate Y-axis direction

Step 5 Make the tool side of the sensor parallel to the ground ($J4=0^\circ$), as shown in Figure 4.5.

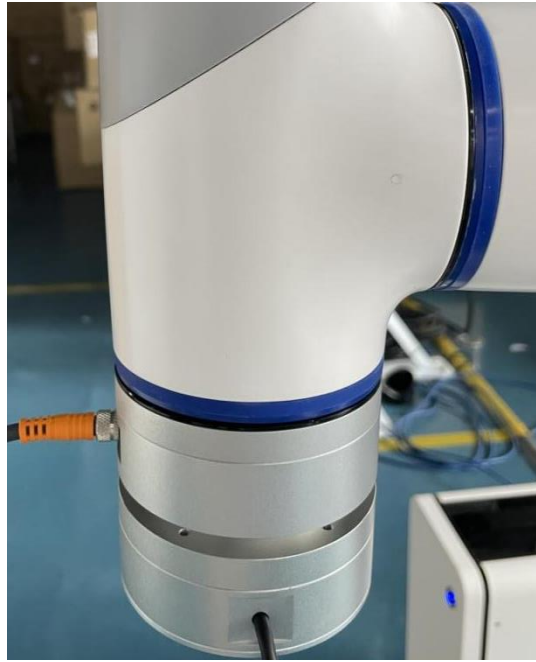


Figure 4.5 Sensor parallel to the ground

Step 6 Click **Continue** in the Setting page to calibrate Z-axis direction, as shown in Figure 4.6.

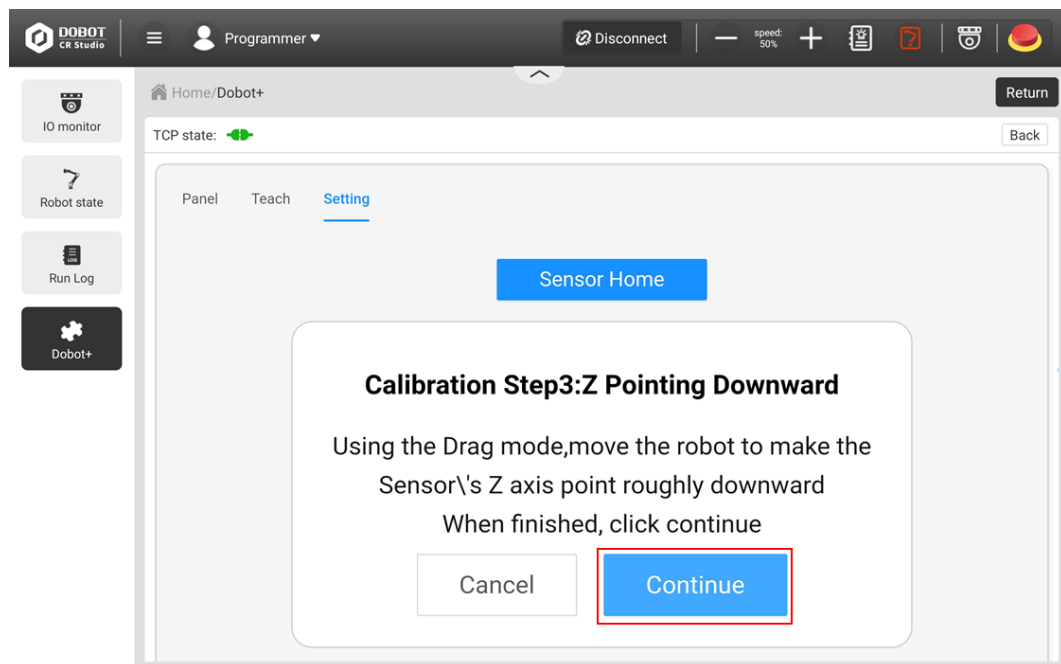


Figure 4.6 Calibrate Z-axis direction

Step 7 After the X, Y and Z axes are calibrated, the screen displays “Calibrate finish”, as shown in Figure 4.7.

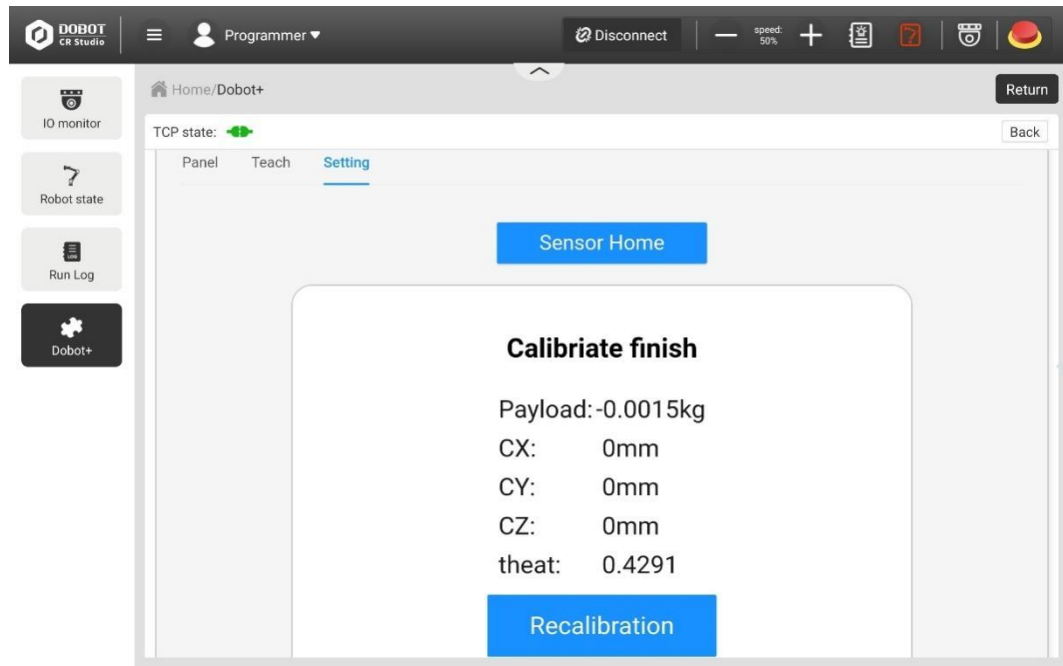


Figure 4.7 Calibration is finished

4.2 Homing the sensor

Before calibrating the sensor, you need to click **Sensor Home** on “Setting” page to reset the sensor, as shown in Figure 4.8.

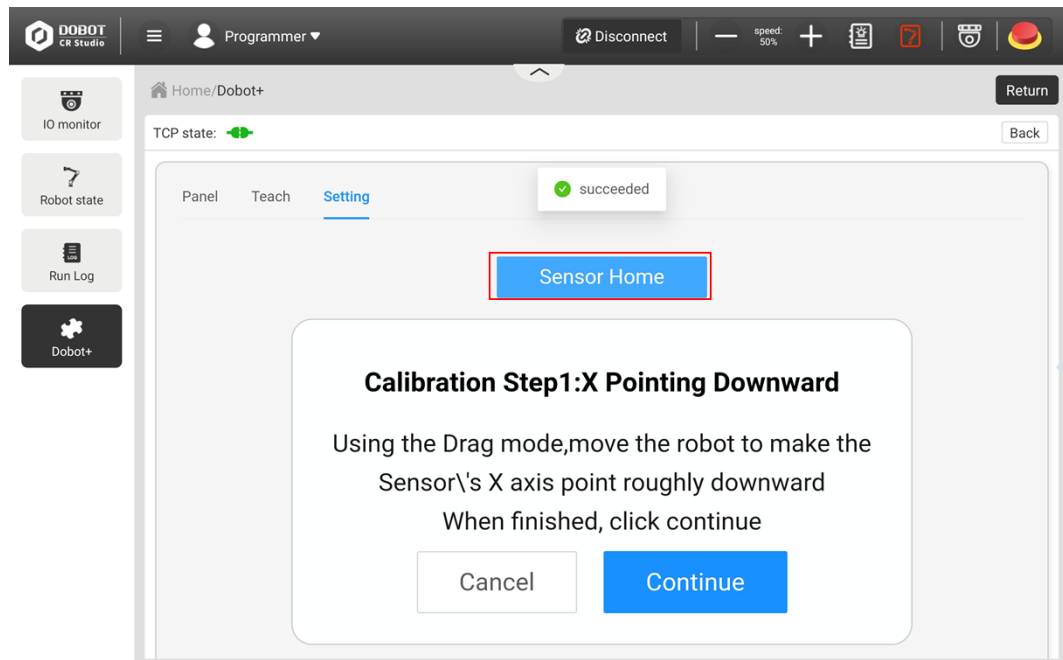


Figure 4.8 Homing the sensor

4.3 Dragging mode

You can enter the “Teach” page to select the dragging mode and adjust the dragging speed, as shown in Figure 4.9.

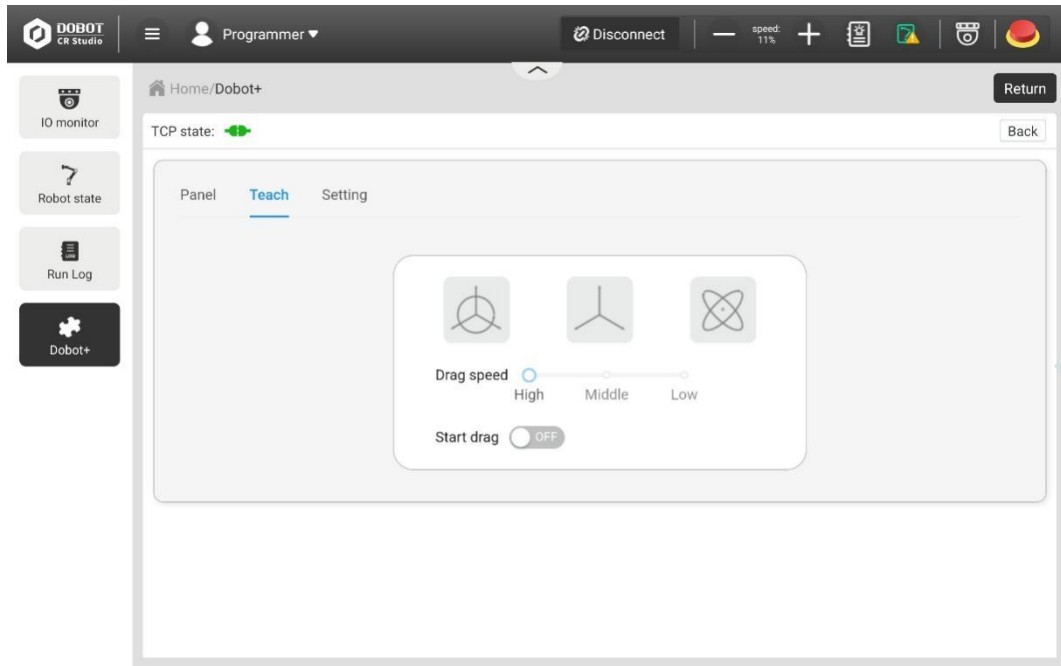


Figure 4.9 Set dragging mode and speed



: Full drag. Change the posture and position of the end tool according to the force on the end.



: Moving drag. The robot changes its position according to the value change of the sensor. The posture of the end tool of the robot does not change during the movement.



: Pose drag. The position of the end tool does not change, while the posture of the end tool changes.

NOTICE

It is recommended to drag with a tool at the end of the robot. You need to grab the bottom of the sensor when dragging without a tool. Holding the side of the sensor does not work!